

Activation of Cyclin D1 by Estradiol and Spermine in MCF-7 Breast Cancer Cells: A Mechanism Involving the p38 MAP Kinase and Phosphorylation of ATF-2

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Estradiol (E₂) and the naturally occurring polyamines (putrescine, spermidine, and spermine) play important roles in breast cancer cell growth and differentiation. We examined the effects of E₂ and spermine on the phosphorylation and DNA binding of activating transcription factor-2 (ATF-2) in MCF-7 breast cancer cells. ATF-2 is a transcription factor involved in estrogenic regulation of cyclin D1 gene, and thereby cell cycle progression. DNA affinity immunoblot assays showed a six- to eightfold increase in the binding of ATF-2 to a 74-mer ATF/CRE oligonucleotide (ODN1) from cyclin D1 promoter in the presence of 4 nM E₂ and 0.5 mM spermine, compared to untreated control. Individual treatments with E₂ or spermine caused a twofold or lower increase in ATF-2 binding to ODN1. Immunoblotting with phospho-ATF-2 antibody showed that increased DNA binding of ATF-2 was associated with its phosphorylation. A p38 MAP kinase inhibitor, PD169316, inhibited ATF-2 phosphorylation. In contrast, the MEK-ERK1/2 inhibitor, PD98059, or the JNK inhibitor, SP600125, had no significant effect on DNA binding of ATF-2. Cyclin D1 promoter (–1745CD1) activity increased by approximately 12-fold (above control) in the presence of E₂ and spermine, compared to a sixfold increase in the presence of E₂ alone and a twofold increase with spermine. Cells transfected with a dominant negative mutant of ATF-2 showed decreased transactivation of cyclin D1 promoter in response to E₂ and spermine. These results indicate that spermine can enhance E₂-induced cell signaling and cyclin D1 transcription by activation of the p38 MAP kinase and phosphorylation of ATF-2, contributing to breast cancer cell proliferation.

Key words: ATF-2; Cyclin D1; Estradiol; Polyamines; MCF-7 cells

Estradiol (E₂)-induced breast cancer cell growth involves activation of cyclin D1, an important cell cycle regulatory protein (1,2). E₂ generally acts through ligand-activated transcription factors, estrogen receptor α and β (ER α and ER β), which bind to estrogen response elements (ERE) in the promoter region of estrogen responsive genes (3,4). However, the mechanism of estrogenic signaling, leading to the activation of cyclin D1, is complicated by the lack of an ERE sequence in the

promoter region of this gene (2,5). It has been suggested that cyclin D1 transcription involves the cooperation of multiple signal transduction pathways and transcription factors (6,7). The promoter region of cyclin D1 contains binding sites for AP-1, SP1, NF- κ B, STAT, and ATF/CREB (6,7). Among these sites, the ATF/CREB binding site has a unique role in E₂-induced transcriptional activation of the cyclin D1 gene in MCF-7 breast cancer cells (5). The ATF-2 protein was required for E₂-induced

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transcriptional activation of the cyclin D1 promoter; however, the mechanism(s) by which E_2 facilitates this process is not known.

ATF-2 is a member of the ATF/CREB protein family of transcription factors (8). ATF-2 contains a leucine zipper motif (bZIP) that forms homodimers and heterodimers with other members of the ATF family, as well as with members of the Jun/Fos family (9,10). Following dimerization, ATF-2 recognizes and binds to ATF/CRE motifs (TGACGTCA) and regulates transcription of genes, including c-Jun (10), TNF- α , TGF- β , cyclin A (13), and cyclin D1 (14). ATF-2 is regulated posttranslationally by phosphorylation, in part by the JNK/SAPK and the p38 group of MAP kinases (15–18). In breast cancer cells, cooperation between ER and ATF-2 might lead to transcriptional activation of cyclin D1. Previous studies from our laboratory and others suggested a link between estrogenic and polyamine pathways in breast cancer cells (19,20). We hypothesized that polyamines might be involved in the cooperative signaling mechanism involving ER α and ATF-2 in breast cancer cells.

Polyamines—putrescine [$H_2N(CH_2)_4NH_2$], spermidine [$H_2N(CH_2)_4NH(CH_2)_3NH_2$], and spermine [$H_2N(CH_2)_3NH(CH_2)_4NH(CH_2)_3NH_2$]—are organic cations with multiple functions in cell growth and differentiation (21,22). Polyamine levels are elevated in breast tumors compared to adjacent normal tissues (23). Elevated levels of polyamines modify many cellular functions; however, the mechanism(s) by which polyamines stimulate cell growth is not well defined. One possibility is that polyamines alter signaling involving kinase cascades (24–26). For example, the activity of casein kinase 2 is stimulated by the interaction of spermine with the β subunit of the enzyme (25). Polyamines are also reported to modulate specific protein–protein interactions, involving ER and its coactivators/corepressors, and vitamin D receptor and its interacting proteins, DRIP205 and GRIPI (27,28). At the level of gene expression, polyamines have regulatory functions at the transcriptional and posttranscriptional levels. Transcriptional regulation of *c-myc* gene by polyamines was first demonstrated by the use of inhibitors of polyamine biosynthesis (29). Ornithine decarboxylase (the first rate-limiting enzyme in polyamine biosynthesis) is also a well-defined target of *c-myc* (30). Microarray studies on *E. coli* showed upregulation of 309 out of 2742 genes by polyamines, mainly at the translational level (31). The critical role of polyamine pathway in malignancy is provided by a recent report of meta-analysis of four independent microarray studies (32). Aberrations in polyamine pathway and purine metabolism were the most consistent observations in the meta-analysis of studies using oligonucleotide and cDNA arrays, indicating a fundamental role for polyamine pathway in the dysregulation of gene expression in malignancy.

While it is desirable to map out all polyamine-modulated signaling pathways, unraveling the components of the linkage between estrogenic and polyamine pathways in breast cancer is essential to develop therapeutic strategies involving these pathways. In the present study, we examined the effects of E_2 and spermine on ATF-2 phosphorylation and the binding of phosphorylated ATF-2 to cyclin D1 promoter DNA. Our results show that E_2 and spermine can exert a synergistic increase in ATF-2 binding to cyclin D1 promoter. The facilitation of ATF-2 binding to DNA is associated with ATF-2 phosphorylation by the p38 MAP kinase, and increased cyclin D1 transactivation.

MATERIALS AND METHODS

Chemicals, Reagents, and Antibodies

17 β -Estradiol was purchased from Steraloids, Inc. (Wilton, NH). Putrescine, spermidine, and spermine were obtained from Sigma Chemical Co. (St. Louis, MO). DMEM, phenol red-free DMEM, and fetal bovine serum were obtained from Sigma. Antibiotics, trypsin, and other additives for cell culture medium were purchased from Gibco Laboratories (Grand Island, NY). Dynabeads M-280 Streptavidin and Dynal MPC-S were obtained from Dynal Biotechnology, Inc. (Lake Success, NY). Biotin-16-2'-deoxyuridine-5'-triphosphate (Biotin-16-dUTP) was purchased from Sigma. HexaPlus DNA labeling kit was obtained from MBI Fermentas (Amherst, NY). Anti-ATF-1, ATF-2, phospho-ATF-2, CREB, ER α , c-Fos, c-Jun, NF- κ Bp50, NF- κ Bp65, p38, and phospho-p38, as well as HRP-conjugated IgG antibodies were obtained from Santa Cruz Biotechnology, Inc. (Santa Cruz, CA). The multicolored protein markers were obtained from NEN Life Science Products (Boston, MA). Inhibitors of p38 MAP kinase (PD169316), ERK1/2 (PD98059), and JNK (SP600125) were obtained from CalBiochem Corp. (San Diego, CA).

Plasmids

The cyclin D1 luciferase reporter plasmids –1745C D1LUC, –1745ATF/CREmutLUC, and the control plasmid pA₃LUC were described previously (33,34). A mutant human ATF-2 complementary DNA (encoding Thr⁶⁹ and Thr⁷¹ substitutions to alanine), subcloned into the eukaryotic expression vector, pCMV, was used to express ATF-2 dominant negative mutant in transfected cells (35). The pRL-SV40 vector (Promega, Madison, WI), containing the *Renilla* luciferase gene was used as an internal control for transfection efficiency in all experiments.

Oligonucleotides

HPLC-purified ODNs were purchased from Oligos, Etc., Inc. (Wilsonville, OR). Single stranded 74-mer

ODNs (−78 to −5) were annealed to prepare the 74-base pair (bp) double-stranded ODN1. The consensus ATF/CRE (underlined) and NF-κB response element (bold) are present in ODN1. A mutated CRE/ATF ODN (ODN2), with a 3-bp change (bold and underlined), and a mutated NF-κB ODN (ODN3), with a 5-bp change (bold and underlined), was also used. The base sequence of the top and bottom strands of the 74-mer ODNs are given: ODN1: 5′-GGGCTTTGATCTTTGCTTAACAACAGTAACGTCACACGGACTACAGGGGAGTTTGTGTTGAAGTTGCAAAGTCCT-3′ and 3′-CCCGAAACTAGAAACGAATTGTTGTCATTGCAGTGTGCCTGATGTCCCTCAAAACAACCTTCAACGTTTCAGGA-5′; ODN2: 5′-GGGCTTTGATCTTTGCTTAACAACACTAGCGGCACACGGACTACAGGGGAGTTTGTGTTGAAGTTGCAAAGTCCT-3′ and 3′-CCCGAAACTAGAAACGAATTGTTGTTGATCGCCGTGTGCCTGATGTCCCTCAAAACAACCTTCAACGTTTCAGGA-5′; ODN3: 5′-GGGCTTTGATCTTTGCTTAACAACAGTAACGTCACACGGACTACAGATGCGTGCTGTTGAAGTTGCAAAGTCCT-3′ and 3′-CCCGAAACTAGAAACGAATTGTGTCATTGCAGTGTGCCTGATGTCTACGCACGACAACCTTCAACGTTTCAGGA-5′.

Cell Culture and Transient Transfection Assay

MCF-7 breast cancer cells were obtained from the American Type Culture Collection (Rockville, MD). Cells were maintained in DMEM, and supplemented with 100 U/ml penicillin, 100 µg/ml streptomycin, 40 µg/ml gentamicin, 2 µg/ml insulin, 0.5 mM sodium pyruvate, 50 µM nonessential amino acids, 2 mM L-glutamine, and 10% fetal bovine serum (36,37). Ten days prior to each experiment, MCF-7 cells were grown in DMEM without phenol red and containing 10% dextran-coated, charcoal-treated serum (36,37). All experiments with exogenous polyamines were conducted in the presence of 1 mM aminoguanidine to inhibit their oxidation by serum-derived polyamine oxidases. For transfection assay, 3.5×10^5 cells were plated in 12-well microtiter plates for 24 h. Using the calcium phosphate coprecipitation procedure, cells were transfected with 1 µg of cyclin D1 promoter plasmids linked to a luciferase reporter gene (i.e., −1745 CD1LUC, −1745ATF/CREmutLUC) in 50 µl of transfection buffer. In other experiments, cells were also transfected with 10 µg of ATF-2 dominant negative mutant expression vector to test the effects of ATF-2 dominant negative mutation on cyclin D1 transactivation. Twenty-four hours after transfection, cells were treated in triplicate with 4 nM E_2 , 0.1 to 1 mM spermine, or E_2 and spermine combination for 7 h. The activities of the cyclin D1 firefly reporter gene and the *Renilla* luciferase internal control were determined by a Dual Luciferase reporter assay system (Promega). Corrections were

made for transfection efficiency and variations in harvesting by cotransfection with *Renilla* luciferase reporter gene driven by a SV40 promoter. Reporter activity was normalized for each sample using the following equation: normalized luciferase activity = firefly luciferase activity/*Renilla* luciferase activity.

Cellular Extract Preparation and DNA Affinity Immunoblot Assay

MCF-7 cells (3.5×10^6 /100 mm dish) were treated with 4 nM E_2 or 0.1, 0.25, 0.5, and 1 mM of spermine, or 4 nM E_2 in combination with 0.1 to 1 mM spermine for 7 h. Cells were harvested in PBS and sonicated in lysis buffer [150 mM Tris-HCl (pH 7.4), 150 mM NaCl, 1% Nonidet P-40, 2 mM EDTA, 50 mM sodium fluoride, 0.2% SDS, 1 mM sodium vanadate, 2 µg/ml leupeptin, aprotinin, and pepstatin, and 1 mM phenylmethylsulfonyl fluoride]. Cell lysate was cleared of cellular debris by centrifugation at $12,000 \times g$ for 15 min and used for DNA affinity immunoblot assay.

We used DNA affinity beads coated with streptavidin (Dynabeads) and biotin 5′ end-labeled cyclin D1 ODNs containing the ATF/CRE and NRE consensus sequences in our affinity assays (38,39). ODNs (200 nM) were biotinylated using the HexaLabel Plus DNA labeling kit. Biotinylated ODNs (2 nM) were incubated with Dynabeads (1 mg/ml) at 22°C in TE buffer containing 1 M NaCl for 30 min with gentle rotation. The Dynabeads containing the immobilized biotinylated DNA were washed, separated in a DYNAL MPC-S, and incubated with cellular extract (75–100 µg protein) in 8 mM Tris-phosphate at pH 7.4, 0.12 M KCl, 8% glycerol, 4 mM DTT, and 0.5% CHAPS for 1 h at 4°C. Subsequently, beads were washed in 10 mM HEPES at pH 7.7, 50 mM KCl, 20% glycerol, and 0.1% NP-40. Dynabeads were boiled in 100 µl Laemmli buffer, and DNA-bound proteins separated on 10% polyacrylamide gel and identified by immunoblotting with specific antibodies. To ensure equal loading, blots were stripped and reprobed with antibody to ERα or p50, which did not change with treatments.

Western Blot Analysis

Cell lysate was prepared according to the procedure previously described (33,36). Briefly, monolayers of MCF-7 cells (3.5×10^6 /100 mm dish) were washed twice with ice-cold PBS and lysed by the addition of ice-cold lysis buffer [150 mM Tris-HCl (pH 7.4), 150 mM NaCl, 1% Nonidet P-40, 2 mM EDTA, 50 mM sodium fluoride, 0.2% SDS, 1 mM sodium vanadate, 2 µg/ml leupeptin, aprotinin, and pepstatin, and 1 mM phenylmethylsulfonyl fluoride]. After 30 min on ice, lysate was cleared of cellular debris by centrifugation ($15,000 \times g$) for 20 min at 4°C and the supernatant frozen at −20°C

in aliquots. Protein (30 μ g) (determined by the Bradford protein assay) was diluted in 2 $\times$ SDS-PAGE Laemmli buffer [150 mM Tris base (pH 6.8), 30% glycerol, 4% SDS, 7.5 mM dithiothreitol, 0.01% bromophenol blue] and separated on a 10% SDS-polyacrylamide gel. After electrophoresis, separated proteins were transferred to PVDF Polyscreen membrane. Membrane was incubated in 5% nonfat milk in Tris-buffered saline, containing 0.1% Tween-20 for 1 h to saturate the nonspecific binding sites. Membrane was then immunoblotted with a 1:200 dilution of the primary antibody. Levels of p38 MAPK (total) and phospho-p38 MAPK (activated) were detected with anti-p38 and anti-phospho-p38 MAPK monoclonal antibodies, respectively. Protein bands were visualized using horseradish peroxidase-conjugated secondary antibody with a chemiluminescence-based detection system. To verify equal protein loading, membranes were stripped and reblotted with anti- β -actin monoclonal antibody (1:5000). Intensity of protein bands was quantified using a Scanjet 4C flatbed scanner (Hewlett-Packard) with NIH Image v1.52 software. Lightly exposed films were used for quantification.

HPLC Analysis of Intracellular Polyamines

To determine intracellular polyamine levels, MCF-7 cells were plated in 100-mm dishes and treated with various concentrations of spermine, spermidine, or putrescine in the presence or absence of 4 nM E_2 . The cell pellet was sonicated in 300 μ l of 8% sulfosalicylic acid on ice. The solution was incubated in ice for 1 h and centrifuged to remove the precipitated protein. After precolumn dansylation, intracellular concentrations of spermine, spermidine, and putrescine were quantified by a previously described procedure (37), using a Hitachi D-7000 HPLC unit. 1,6-Diaminohexane was used as the internal standard.

[3 H]Thymidine Incorporation Assay and Cell Proliferation

MCF-7 cells (0.5×10^6) were seeded in six-well plates in phenol red-free DMEM supplemented with 10% DCC-treated serum and additives. After 24 h, cells were treated for 24 h with 4 nM E_2 and 0.5 mM spermine, as single agents, or in combination. DNA synthesis was measured by adding 4 μ Ci/ml of [3 H]thymidine to cells 1 h prior to the specified time point. Cells were washed twice with ice-cold PBS and ice-cold 5% trichloroacetic acid. The cell layer was then solubilized in 1 N NaOH and neutralized with 1 N HCl. The radioactive thymidine incorporated to cellular DNA was quantified by liquid scintillation counting. For the determination of cell number, cells were plated and treated in the same manner as [3 H]thymidine incorporation. Cells were harvested

by trypsinization and live cells were counted by trypan blue exclusion, using a hemocytometer.

Statistical Analysis

Results of transient transfection studies are presented as means $\pm$ SEM, for at least three separate experiments. All blots shown are representative of three to four separate experiments with mean of fold changes in intensities reported in the text. Statistical difference between control and treatment groups was determined by one-way analysis of variance (ANOVA) followed by Dunnet's test (GraphPad Prism Software program, San Diego, CA). A value of $P < 0.05$ was considered to be statistically significant.

RESULTS

Polyamines Enhance the DNA Binding and Phosphorylation of ATF-2 in MCF-7 Breast Cancer Cells in the Presence of Estradiol

We examined the effects of E_2 and spermine on the binding of ATF-2 to a 74-mer oligonucleotide (ODN1) from the estrogen-responsive region of the cyclin D1 promoter, containing an ATF/CRE motif. Cells were treated with 4 nM E_2 , 0.5 mM spermine, or E_2 + spermine for 7 h. This time point was chosen based on our cyclin D1 transactivation data, which showed maximal induction of cyclin D1 promoter activity at 7 h. Cellular extracts were prepared and binding to ODN1 determined by DNA affinity immunoblot assay (39). A combination of E_2 and spermine induced a significant eightfold increase ($P < 0.0001$; $n = 4$) in the binding level of ATF-2 to ODN1 compared to the control (Fig. 1A). In contrast, ATF-2 binding showed less than twofold increase in the presence of 4 nM E_2 or 0.5 mM spermine. To determine whether the increased level of binding of ATF-2 to ODN1 was associated with its phosphorylation, we conducted DNA affinity immunoblot assays using an anti-phospho-ATF-2 antibody. A sixfold increase in the level of phosphorylated ATF-2 was detected in cells treated with E_2 and spermine, compared to the control. The increase in DNA binding of ATF-2 was not a result of increased protein abundance because Western blot analysis showed a comparable level of the ATF-2 protein in both treated and untreated samples (Fig. 1A).

We also examined the level of ER α association with ODN1 during the DNA affinity immunoblot assay. Although ODN1 does not contain a classical ERE, ER α may interact as part of a multiprotein complex. As shown in Figure 1A, ER α was detected in association with ODN1 by DNA affinity immunoblot assay, and this association was not modified by E_2 or spermine as single agents or in combination. Thus, ER α can be used as an

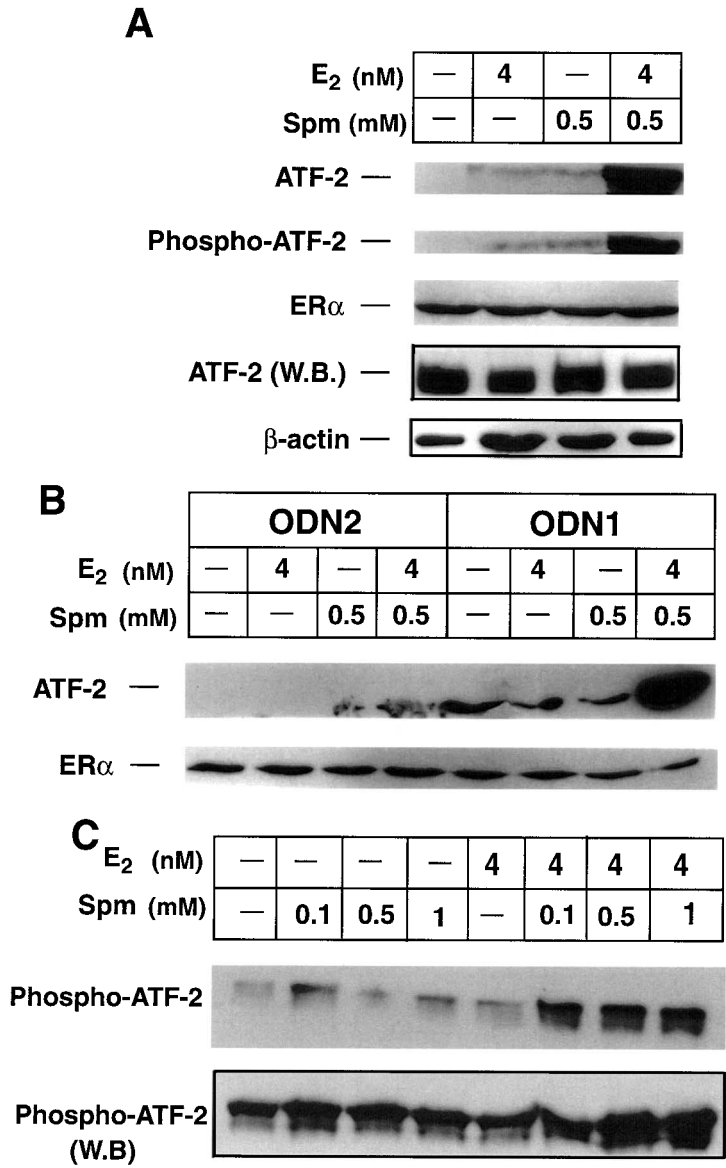


Figure 1. (A) Effects of E₂ and spermine (Spm) on the binding of ATF-2 to the CRE/ATF site of cyclin D1 (ODN1). MCF-7 cells were treated with the indicated concentrations of E₂ and spermine for 7 h, and cellular extract analyzed by DNA affinity immunoblot assay using anti-ATF-2, anti-phospho-ATF-2, or anti-ER antibodies. Results of direct Western blotting (W.B.) experiments, using anti-ATF-2 and anti- β -actin antibodies, are also shown. (B) DNA affinity immunoblot assay using ODN1 and its mutant, ODN2. Results using anti-ATF-2 and anti-ER α antibodies are shown. (C) Effect of increasing concentrations of spermine on phosphorylation of ATF-2 and its binding to ODN1 in the absence and presence of E₂. DNA affinity immunoblot and direct Western blots were conducted using anti-phospho-ATF-2 antibody. Equal loading of samples in DNA affinity immunoblot and Western blot experiments was confirmed by using anti-ER α and anti- β -actin antibodies, respectively. Results presented are representative of three to four separate experiments. Western blots are marked by solid lines.

internal control for the level of proteins in samples during DNA affinity immunoblot assay.

We further tested the importance of the ATF/CRE motif in ATF-2 binding by performing DNA affinity immunoblot experiments using an oligonucleotide (ODN2) with a 3-bp mutation at the ATF/CRE site. The disruption of the ATF/CRE motif in ODN2 resulted in a major loss of ATF-2 binding in the presence of E_2 and spermine (Fig. 1B). However, there was a minor band of ATF-2 that indicated a low level of protein–DNA binding in the presence of spermine, regardless of the presence of E_2 . A relatively low level of ATF-2 DNA binding was found in some experiments even in the absence of estradiol, although the effects of combinations of estradiol and spermine were six- to eightfold higher than that of individual treatments. ODN2 was capable of retaining the ER binding in our DNA affinity immunoblot assay. This result suggests that alternate ER α binding sites may exist either through protein–protein associations or through nonconventional DNA binding sites (41,42). Overall, these results show that the integrity of the ATF/CRE motif is needed for the increase in ATF-2 binding to ODN1 in the presence of E_2 + spermine.

To understand the dose dependence of spermine's effects on ATF-2 phosphorylation and concomitant increase in DNA binding, cells were treated with increasing concentrations of spermine with or without E_2 . The extent of phosphorylation of ATF-2 was determined by Western blotting using an anti-phospho-ATF-2 antibody and the binding of phosphorylated ATF-2 to DNA was measured by DNA affinity immunoblot assay using ODN1. Our results (Fig. 1C) show that in the absence of E_2 , spermine (0.1, 0.5, and 1 mM) had relatively minor effects on the DNA binding of ATF-2. In contrast, there was a six- to eightfold increase in the level of binding of ATF-2 to ODN1 in the presence of 0.1–1 mM spermine and 4 nM E_2 . Western blotting using anti-phospho-ATF-2 antibody showed no significant change in phosphorylation in the presence of either spermine or E_2 as single agents. However, there was a two- to fourfold increase in the level of phosphorylated ATF-2 in the presence of spermine (0.1–1 mM) and E_2 .

We also examined the effects of putrescine and spermidine on ATF-2 binding to ODN1 in MCF-7 cells growing in the presence of E_2 . Both compounds were less effective compared to spermine in enhancing ATF-2 binding to ODN1, with a maximal increase of twofold over the control, occurring at 2–5 mM concentrations (results not shown). No significant effect was found when putrescine or spermidine was added in the absence of E_2 . These results are consistent with the lower efficacy of putrescine and spermidine compared to spermine, in reactions involving DNA and proteins (22).

To determine if exogenous polyamines were internalized in MCF-7 cells, we quantified intracellular polyamines by HPLC. Treatment of cells with 0.1 mM spermine and 4 nM E_2 resulted in a threefold increase in spermine levels compared to control (Table 1). In the presence of 1 mM exogenous spermine and E_2 , intracellular spermine levels were sixfold higher than that in control cells. Spermine treatment reduced putrescine level by 50% compared to control. In parallel experiments, we determined the effects of putrescine and spermidine on intracellular polyamine levels. Treatment of cells with 1–10 mM putrescine resulted in a 12- to 16-fold increase in intracellular putrescine levels, while treatment with 1–10 mM spermidine caused a two- to fourfold increase in intracellular spermidine levels.

Phosphorylation of ATF-2 by Estradiol and Spermine Involves the p38 MAP Kinase, and not MEK/ERK or JNK/SAPK

To investigate the signal transduction pathways involved in E_2 and spermine-induced phosphorylation of ATF-2 in MCF-7 cells, we next examined whether chemical inhibitors (43,44) that target specific kinases could inhibit ATF-2 phosphorylation. We first confirmed the ability of PD169316, a p38 kinase inhibitor, PD98059, an inhibitor of the MEK/ERK, and SP600125, an inhibitor of JNK/SAPK for their ability to inhibit these kinases in MCF-7 cells. CdCl₂ (10 μ M) induction of these kinases (45) followed by Western blot analysis of phosphorylated forms of p38, ERK, and c-Jun confirmed the activity of the kinase inhibitors.

MCF-7 cells were pretreated with increasing concentrations of PD169316, a p38 kinase inhibitor, for 30 min and then treated with either 4 nM E_2 , 0.5 mM spermine, or E_2 + spermine for 7 h. Cellular extracts were then analyzed by DNA affinity immunoblot assay. Our results showed that DNA binding of phosphorylated ATF-2 was reduced by 60% by 1 μ M PD169316 and to undetectable level at 10 μ M concentration (Fig. 1). Western blot analysis of cell lysate using anti-phospho-ATF-2 antibody showed that 10 μ M PD169316 suppressed the phosphorylation of ATF-2, although ATF-2 DNA binding was more sensitive to the action of the inhibitor than phosphorylation. Western blotting using an anti-ATF-2 antibody demonstrated that total ATF-2 protein levels were not significantly altered by the presence of PD169316. We also examined the effects of inhibitors of the MEK/ERK (PD98059) and JNK/SAPK (SP600125) pathways on the DNA binding of ATF-2. These inhibitors did not significantly affect the DNA binding activity of phosphorylated ATF-2 (Fig. 2B, C).

Table 1. Polyamine Levels in MCF-7 Cells After Treatment With Spermine, Spermidine, and Putrescine in the Absence or Presence of 4 nM E₂ in Culture

E ₂	Treatment/ Concentration (mM)			Cellular Polyamine Concentrations (nmol/mg Protein)		
	SPM	SPD	PUT	PUT	SPD	SPM
Control	—	—	—	2.7 ± 0.5	11.3 ± 1.7	4.6 ± 1.2
—	0.1	—	—	1.8 ± 0.4	9.0 ± 2.0	12.08 ± 2.2
—	0.25	—	—	1.1 ± 0.2	8.6 ± 1.6	16.2 ± 3.3
—	0.5	—	—	1.0 ± 0.3	9.8 ± 2.5	23.7 ± 1.4
—	1.0	—	—	1.1 ± 0.3	10.4 ± 2.0	34.9 ± 5.0
4 nM	—	—	—	2.4 ± 0.8	8.7 ± 2.4	9.3 ± 2.0
4 nM	0.1	—	—	1.3 ± 0.4	9.9 ± 1.3	15.5 ± 1.3
4 nM	0.25	—	—	1.3 ± 0.3	8.9 ± 1.3	21.3 ± 4.3
4 nM	0.50	—	—	1.4 ± 0.3	11.6 ± 3.0	28.5 ± 2.7
4 nM	1.0	—	—	1.2 ± 0.2	11.5 ± 2.3	33.6 ± 3.9
—	—	0.5	—	1.6 ± 0.5	20.7 ± 1.2	6.4 ± 0.9
—	—	1.0	—	1.7 ± 0.1	26.9 ± 0.4	6.5 ± 0.15
—	—	2.5	—	2.0 ± 0.2	31.4 ± 1.2	7.5 ± 1.6
—	—	5.0	—	2.6 ± 1.0	40.1 ± 4.0	10.5 ± 2.8
4 nM	—	0.5	—	3.5 ± 1.7	25.9 ± 4.5	6.0 ± 1.5
4 nM	—	1.0	—	1.9 ± 0.6	28.5 ± 3.6	7.4 ± 1.5
4 nM	—	2.5	—	2.7 ± 1.1	35.9 ± 2.9	8.3 ± 1.8
4 nM	—	5.0	—	3.4 ± 1.2	40.1 ± 2.0	10.5 ± 1.4
—	—	—	1.0	33.5 ± 3.3	6.1 ± 0.4	3.3 ± 0.1
—	—	—	5.0	38.3 ± 1.3	5.5 ± 0.5	3.5 ± 0.6
—	—	—	10.0	46.0 ± 3.0	6.45 ± 1.3	4.3 ± 0.9
4 nM	—	—	1.0	32.5 ± 2.2	5.03 ± 0.1	3.1 ± 0.1
4 nM	—	—	5.0	41.5 ± 3.0	5.9 ± 0.3	3.4 ± 0.6
4 nM	—	—	10.0	45.8 ± 1.4	6.0 ± 0.6	3.9 ± 0.3

Data presented are the mean ± SD of triplicate measurements. Polyamine levels were quantified by the HPLC technique. Cells were treated with spermine, spermidine, or putrescine for 7 h in the absence or presence of 4 nM E₂. SPM, spermine; SPD, spermidine; PUT, putrescine.

Estradiol and Spermine Could Increase Phosphorylation of the p38 MAP Kinase in MCF-7 Cells

We next examined the effect of E₂ + spermine on the activation of p38 MAP kinase by its phosphorylation. Cells were treated with 4 nM E₂, 0.5 mM spermine, or 4 nM E₂ + 0.5 mM spermine combination for 7 h, lysed, and the level of active p38 MAP kinase detected by immunoblotting with an antibody specific to phosphorylated p38. Our results showed that a combination of E₂ and spermine increased phosphorylated p38 level by a significant fivefold compared to untreated control (Fig. 3). Individual treatment of cells with E₂ resulted in threefold increase of phospho-p38 levels whereas spermine treatment led to a twofold increase. These results show that

a combination of E₂ + spermine had an additive effect in activating p38.

Binding of Multiple Transcription Factors to a CRE/ATF Motif in the Presence of Estradiol and Spermine

Members of the CREB/ATF family of proteins bind to DNA as intrafamily heterodimers, as well as cross-family heterodimers with members of AP-1 family (9). Therefore, we next examined the DNA binding activity of ATF-1, CREB, c-Fos, c-Jun, and the NF-κB subunit p50 in MCF-7 cells to test whether the effect of E₂/spermine combination is specific to ATF-2. Treatment of cells with E₂ and spermine caused <40% increase in the binding level of ATF-1 and CREB to ODN1 (Fig. 4).

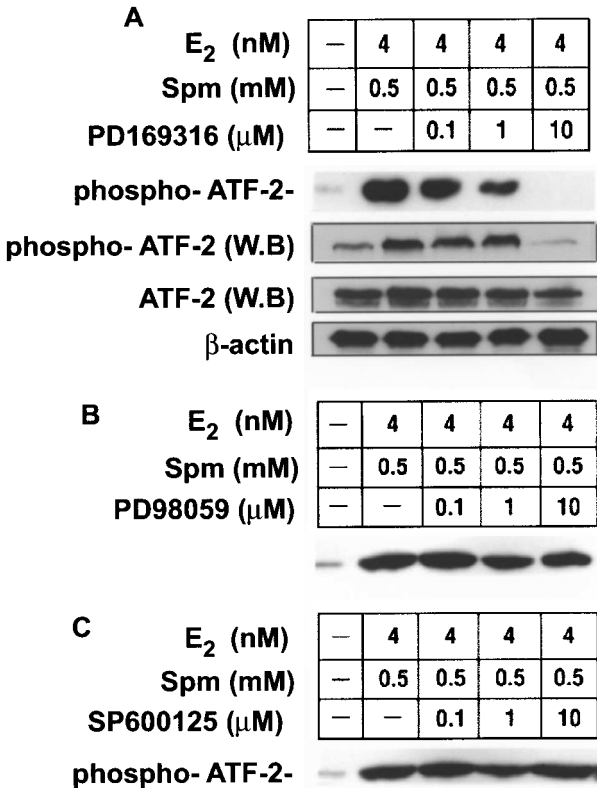


Figure 2. Effects of kinase inhibitors on the binding of phosphorylated ATF-2 to ODN1. (A) MCF-7 cells were treated with indicated concentrations of spermine, E₂, and the p38 MAP kinase inhibitor, PD169316 for 7 h, and DNA affinity immunoblot assay conducted using anti-phospho-ATF-2 antibody. Western blots were performed using anti-phospho-ATF-2, anti-ATF-2, and anti-β-actin antibodies. (B) Effect of the ERK/MEK inhibitor, PD98059, on the binding of ATF-2 to ODN1. DNA affinity immunoblot assay was conducted using anti-phospho-ATF-2 antibody. (C) Effect of the JNK inhibitor, SP600125, on the level of binding of ATF-2 to ODN1. Similar results were obtained in three separate experiments. Western blots are marked by solid lines.

Treatment of cells with combinations of E₂ and spermine increased ATF-1 binding level by 50% and CREB binding level by 80%. Treatment with either E₂ or spermine did not lead to a significant increase in c-Fos and c-Jun binding to ODN1; however, a combination of these agents caused a 2.5-fold (c-Fos) and threefold (c-Jun) increase in their binding to ODN1. However, the binding of NF-κB subunit, p50, was not altered by E₂ or spermine treatments. In separate experiments, we examined the binding of NF-κB subunit p65 to ODN1. In this case also there was no significant change in the level of binding in the presence of E₂, spermine, or their combinations (results not shown). These results show that E₂ and spermine enhance the binding of multiple transcrip-

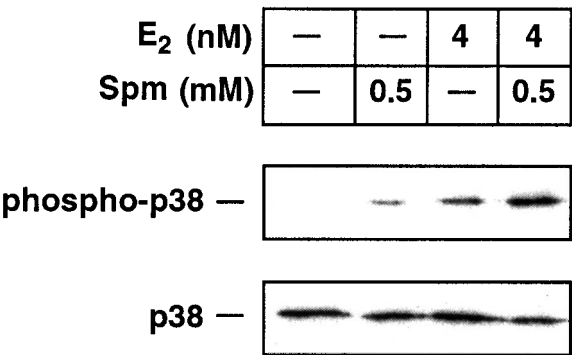


Figure 3. Effects of E₂ and spermine on the phosphorylation of p38 MAP kinase. MCF-7 cells were treated with the indicated concentrations of E₂ and spermine for 7 h, lysed, and active p38 MAPK (phospho-p38) determined by Western blotting technique. The membrane was stripped and reprobed for total p38 MAPK to ensure equal protein loading. Results presented are representative of three separate experiments.

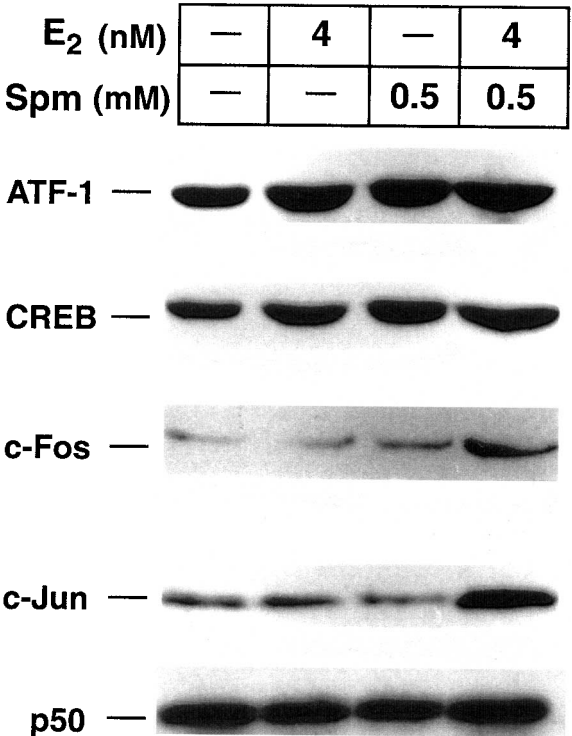


Figure 4. Effect of E₂ and spermine on the binding of ATF-1, CREB, c-Fos, c-Jun, and NF-κB subunit p50 to ODN1. MCF-7 cells were treated with the indicated concentrations of E₂ and spermine for 7 h, and cellular extract analyzed by DNA affinity immunoblot assay. Because the binding of p50 was not altered by treatments in multiple experiments, the level of p50 serves as a control for loading and transfer. Similar results were obtained in two separate experiments.

tion factors, particularly the AP-1 family members to the cyclin D1 promoter region, although these effects are quantitatively less than that of ATF-2 binding. It is to be noted that the DNA affinity immunoblot assay detects the *in vitro* interaction of transcription factors, including those present in the cytosolic fraction, to the cyclin D1 promoter, not the *in vivo* interactions as in the chromatin immunoprecipitation assay. Whole cell extract was used in this assay to include possible binding of nuclear and cytoplasmic proteins, as in some cases nucleocytoplasmic shuttling of ER or other proteins might be affected by the treatments (40).

Effects of Spermine and Estradiol on Cyclin D1 Transcriptional Activation

To test the functional significance of ATF-2 phosphorylation and ATF-2-CRE binding to cyclin D1 promoter in MCF-7 breast cancer cells, we performed transient transfection experiments using a full-length cyclin D1 promoter plasmid linked to the luciferase reporter (–1745CD1LUC). In these experiments, transfected cells were treated with 0.1–1 mM spermine in the absence or presence of 4 nM E₂ for 7 h. The addition of spermine (0.1–1 mM) as a single agent resulted in about twofold increase in cyclin D1 promoter activity in MCF-7 cells (Fig. 5). Addition of E₂ alone resulted in a sixfold increase in the transactivation of cyclin D1 promoter. In the presence of 4 nM E₂ and spermine, a concentration-dependent increase in cyclin D1 promoter activity occurred, with a maximum 12-fold induction compared to control. This increase was significantly higher than that observed with 4 nM E₂ alone. To investigate whether the CRE played a role in synergistic activation of the cyclin D1 gene by E₂ and spermine, we performed transient transfection experiments using a full-length cyclin D1 promoter (–1745ATF/CREmutLUC), containing a mutated ATF/CRE site (TCGCGGTCC compared to TAACGTCA). Mutation of the ATF/CRE site resulted in a 60–70% reduction in cyclin D1 promoter activity induced by E₂ and spermine (Fig. 5B). The increase in –1745 ATF/CREmutLUC promoter activity was 3.4-fold higher than that of control, compared to an 11- to 12-fold increase with the full-length, wild-type promoter. In contrast, a reporter plasmid containing the empty luciferase vector pA₃LUC did not show an increase in the presence of these agents (Fig. 5C). Thus, the activation of the cyclin D1 promoter by E₂ and spermine was specific and the results indicate that ATF-2/CRE site plays a critical role in spermine + E₂-induced cyclin D1 transcription.

We next performed transfection studies with chemical inhibitors that specifically target the ERK, JNK, or p38 MAP kinase pathways. Treatment of cells with the p38 inhibitor PD169316 (10 μ M) reduced E₂/spermine-

induced activation of the cyclin D1 promoter by 50% (Fig. 6). A 75% inhibition of cyclin D1 transactivation was observed at 100 μ M PD169316. These results are consistent with a significant contribution of p38 phosphorylation pathway to E₂/spermine-induced activation of cyclin D1 promoter. In contrast, the JNK and ERK inhibitors, SP600125 and PD98059, reduced cyclin D1 promoter activity by only 8% and 12%, respectively, even at a high (100 μ M) concentration.

Effect of Dominant Negative ATF-2 Mutant on Cyclin D1 Transactivation

We next conducted transfection studies using an ATF-2 dominant negative mutant to verify the importance of ATF-2 in cyclin D1 transactivation. Cells were transfected with the full-length cyclin D1 reporter plasmid along with 1–10 μ g of ATF-2 dominant negative mutant expression vector. Figure 7A shows the dose-dependent effects of dominant negative ATF-2 on transactivation of full-length cyclin D1 promoter in the absence of E₂. The basal level of transactivation (in the absence of E₂) was not strong, and the suppressive effect of dominant negative plasmid was not statistically significant. In the presence of E₂, the dominant negative plasmid downregulated the transactivation of cyclin D1 and the maximal effect was observed with 10 μ g of the plasmid (Fig. 7B). We further examined the effect of dominant negative ATF-2 on cyclin D1 promoter activity in the presence of E₂ and spermine. Transfected cells were treated with 4 nM E₂, 0.5 mM spermine, or E₂ + spermine for 7 h. We found that cells treated with E₂ or E₂ + spermine showed an 8- and 12-fold increase, respectively, in cyclin D1 promoter activity compared to untreated control. The ATF-2 dominant negative mutant significantly reduced the promoter activity (Fig. 7C). Our results also showed that transfection of cells with 10 μ g of the empty vector pCMV5 had no effect on E₂ + spermine-stimulated activation of cyclin D1 (data not shown). These results indicate that the induction of cyclin D1 promoter by E₂ + spermine requires the presence of an intact ATF-2 protein.

Effects of Estradiol and Spermine on Cyclin D1 Expression and Cell Proliferation in MCF-7 Cells

We next examined the effects of E₂ and spermine on the expression of endogenous cyclin D1. As shown in Figure 8A, addition of spermine or E₂ caused a 1.5- or twofold increase in cyclin D1 levels compared to the control. The presence of spermine + E₂ caused a threefold increase in the level of cyclin D1 protein. We also measured DNA synthesis and cell growth in treated cells to determine whether a correlation exists between ATF-2 activation and cell proliferation. MCF-7 cells were

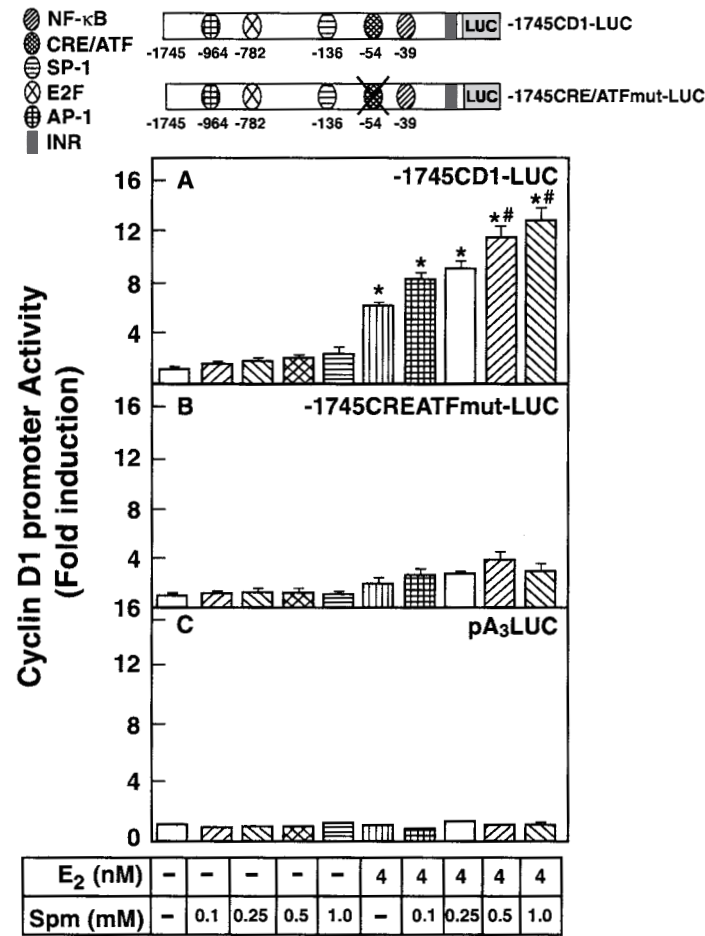


Figure 5. (A) Effects of E₂ and spermine on cyclin D1 promoter activity. MCF-7 cells were cotransfected with a wild-type cyclin D1 plasmid (–1745CD1LUC) and *Renilla* luciferase control plasmid. After 24 h, cells were treated with the indicated concentrations of E₂ and spermine for 7 h, and luciferase activity determined. (B) Effect of ATF/CRE mutation on cyclin D1 promoter activity in the presence of E₂ and spermine. (C) Effects of E₂ and spermine on the induction of luciferase activity from control vector pA₃LUC. Results are shown as mean ± SEM of three to four separate experiments. *Significant (*P* < 0.001) induction compared to untreated control. #Significant (*P* < 0.05) difference between E₂-treated and E₂ + spermine-treated samples.

treated with either 4 nM E₂, 0.5 mM spermine, or 4 nM E₂ + 0.5 mM spermine for 24 h. Cells were harvested and analyzed by [³H]thymidine incorporation assay. Our results showed that treatment of cells with a combination of E₂ and spermine resulted in a 5.5-fold increase in DNA synthesis compared to untreated control cells (Fig. 8B). In contrast, individual treatment with either spermine or E₂ caused a 1.5- and 3.5-fold increase, respectively, in DNA synthesis. In addition, there was a significant increase in cell number after treatment of cells with E₂ + spermine, compared to the control group or the individual E₂/spermine treatment groups (Fig. 8C). Thus, the effect of E₂ and spermine on the stimula-

tion of cyclin D1 gene expression is reflected in cell proliferation.

DISCUSSION

Results presented in this report identify a unique mechanism for enhancing estrogenic signal by spermine in ER-positive MCF-7 cells. This mechanism involves a cell signaling event, leading to the activation of p38 MAP kinase, phosphorylation of ATF-2, and the binding of phosphorylated ATF-2 to cyclin D1 promoter to initiate transcriptional activation of cyclin D1. The cell signaling event might be augmented by the ability of sper-

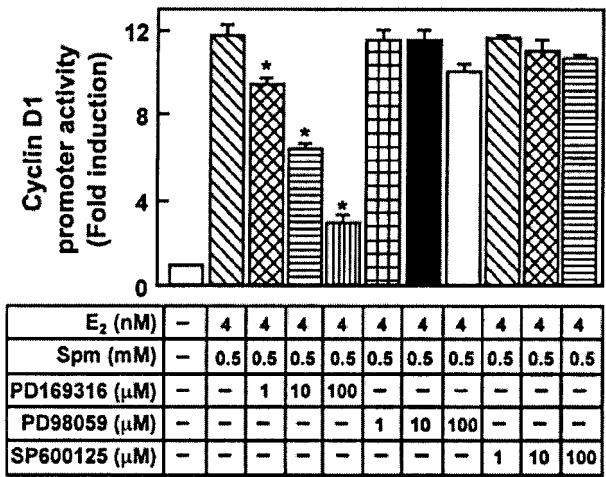
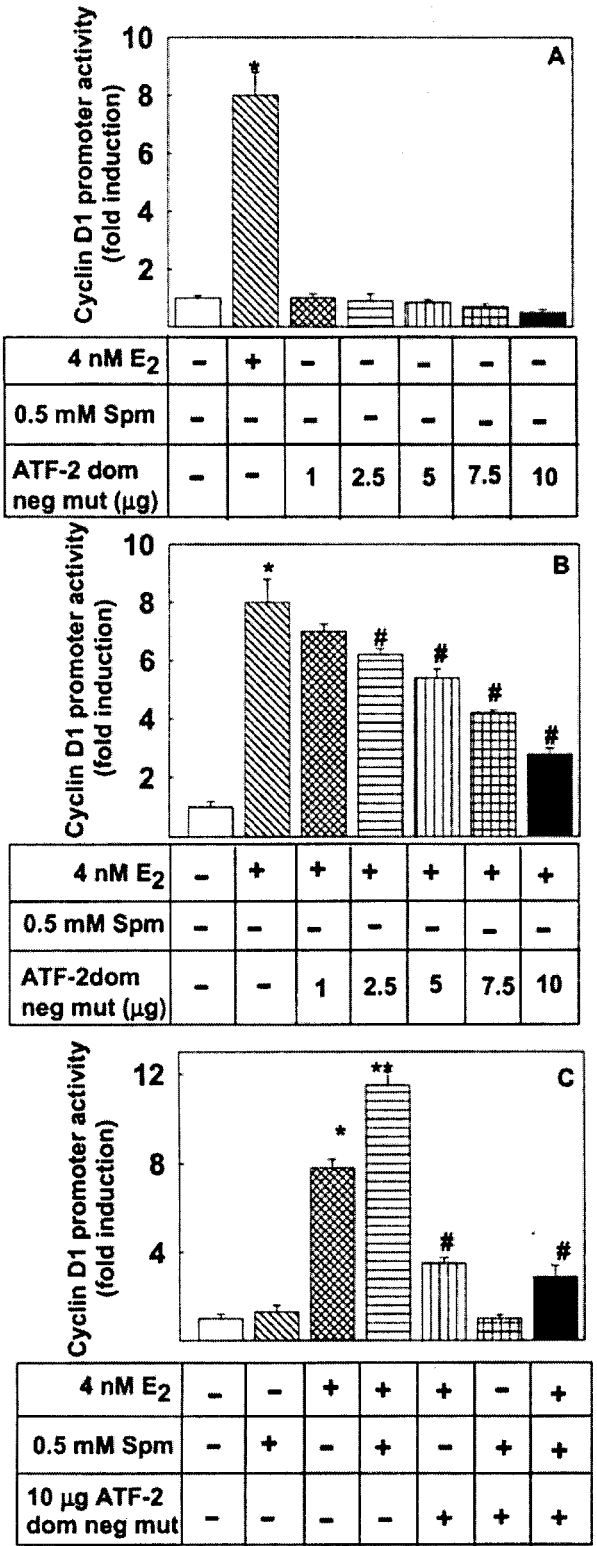


Figure 6. Effects of kinase inhibitors on cyclin D1 promoter activity. MCF-7 cells were cotransfected with a wild-type cyclin D1 promoter plasmid (-1745CD1LUC) and the *Renilla* luciferase control plasmid. Cells were treated with E₂ + spermine (Spm) or E₂ + Spm with one of the inhibitors for 7 h. The kinase inhibitors were added 30 min prior to the addition of E₂ + Spm. *Significant ($P < 0.05$, $n = 3$) inhibition compared to maximal induction in the presence of E₂ + Spm.

mine to facilitate protein–DNA interactions, contributing to the increase in the transactivation of cyclin D1. Spermine also stabilized DNA–protein interactions involving cyclin D1 promoter and c-Fos and c-Jun proteins in the presence of E₂, although this effect was much lower than that of ATF-2. In contrast, there was no significant change in the binding of ATF-1 or p50/p65 to cyclin D1 promoter. Specific transcription factor–DNA interactions might be facilitated by spermine-induced conformational changes in DNA and the assembly of multiprotein complexes on cyclin D1 promoter (36,46–49). Spermine-induced modulation of DNA binding of AP-1, CREB, and other transcription factor binding to their regulatory sequences has been reported (47,48).

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Figure 7. Effect of an ATF-2 dominant negative mutant (dom neg mutant) on cyclin D1 transcriptional activity. (A) Dose dependence of ATF-2 dominant negative mutant on the transactivation of cyclin D1 in the absence of E₂. (B) Dose-dependent effect of dominant negative ATF-2 on the transactivation of cyclin D1 in the presence of E₂. (C) Effect of 10 μg of ATF-2 dominant negative mutant plasmid on transactivation of cyclin D1 in the presence of E₂ and spermine. Cells were treated with 4 nM E₂, 0.5 mM spermine, or E₂ + spermine for 7 h. *Significant ($P < 0.001$, $n = 3$) induction compared to untreated samples. #Significant inhibition ($P < 0.05$) compared to E₂-treated samples. **Significant ($P < 0.05$) induction compared to E₂-treated samples.



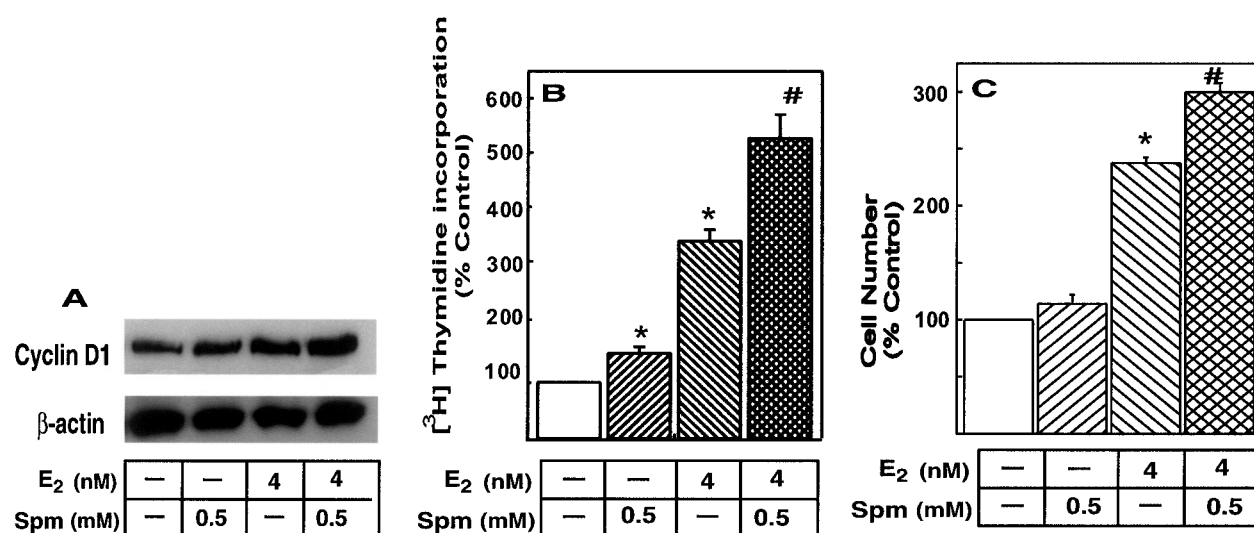


Figure 8. Effect of exogenous spermine on indicators of cell growth in MCF-7 cells. (A) Effect of E₂, spermine, and E₂ + spermine on endogenous cyclin D1 protein levels in MCF-7 cells. Cells were treated with E₂, spermine, or their combinations for 7 h and harvested. Cell lysate was analyzed by Western blots using anti-cyclin D1 and anti-β-actin antibody. (B) Effect of exogenous spermine on DNA synthesis. Cells were treated with 4 nM E₂, 0.5 mM spermine, or E₂ + spermine for 24 h and [³H]thymidine incorporation was determined. (Average CPM for control cells was in the range of 52,000–60,000). (C) Cell number determined by trypan blue exclusion assay. *Significant difference compared to control ($P < 0.05$ ($n = 3$)) and #significant induction compared to E₂-treated samples $P < 0.05$ ($n = 3$).

Spermine-induced activation of the cyclin D1 promoter is mediated by the CRE motif because mutation of this site resulted in a significant reduction of E₂ and spermine responsiveness. This is consistent with the loss of ATF-2 binding to the cyclin D1 promoter, when the CRE site is mutated. Our results further reveal the role of p38 MAP kinase in phosphorylating ATF-2 in the presence of E₂ and spermine in breast cancer cells. In previous studies, Lee et al. (35) reported the involvement of the p38 MAP kinase pathway in the induction of cyclin D1 by the protooncogene pp60^{v-src} in MCF-7 breast cancer cells. Optimal induction of cyclin D1 by pp60^{v-src}, however, required collaborative interactions between the JNK, ERK, and p38 MAP kinase pathways. Phosphorylation of ATF-2 increases its transcriptional activity by relieving intramolecular inhibitory interaction between the DNA binding domain and the transactivating domain (15,18). However, DNA binding and transactivation in the presence of spermine can also be influenced by the stabilization effect of spermine on protein–DNA binding, in addition to the increase mediated by phosphorylation of ATF-2. The high sensitivity of DNA binding of ATF-2 to p38 kinase inhibitor suggests that factors other than ATF-2 phosphorylation cannot be ruled out in the cumulative effects of inhibition of p38 signaling on cyclin D1 promoter.

There has been recent evidence to suggest that E₂ can act via a nongenomic signaling pathway to stimulate the

growth of breast cancer cells (50,51). The nongenomic action of estrogens appears to be mediated by plasma membrane ER, and includes rapid and transient activation of the MAPKs, ERK-1, and ERK-2 (52–56). Other novel pathways for E₂ action include its interaction with the G protein-coupled receptor, GPCR30, which stimulates adenylyl cyclase activity (54) and the stimulation of the p85-associated phosphatidylinositol (PI) 3-kinase/Akt pathway (55). E₂-induced activation of PI3-kinase/Akt pathway was observed in MCF-7 cells and was associated with cyclin D1 activation (55). Akt may activate p38 MAP kinase activity (56); hence, it is possible that E₂ might utilize these alternate signaling pathways to mediate its effect on p38. Studies on the role of polyamines on protein–protein associations, involving ER or vitamin D receptor, suggest a possible redistribution of protein–protein associations in the presence of spermine (27,28). Polyamines may also affect p38 signaling, independent of ER, as spermine has been shown to induce p38 via the Ca²⁺ signaling pathway in an osteoblastic cell line (57). A recent study on a spermine analogue, bis(ethyl)norspermine, showed that prolonged treatment with this compound could result in the phosphorylation of MAP kinases, including p38 (58). In our studies, a combination of E₂ and spermine caused maximal activation of the p38 MAP kinase.

Previous studies by Sabbah et al. (5) have shown an interaction of ER with JUN and ATF-2 proteins using

GST-ER pull-down assay. Our results and those of Sabbha et al. (5) indicate the requirement for JUN/ATF-2 binding site in the promoter for maximal E_2 -induced transactivation of cyclin D1. This result can be interpreted as evidence for ER function exclusively throughc-JUN/ATF-2 complex. However, because DNA affinity immunoblot assay includes the DNA component along with other proteins that might bind to 74-mer ODN, the assay is more likely to detect cumulative DNA–protein and protein–protein interactions. Our results indicate ER α to be a part of the multiprotein complex formed by protein–protein interactions on the cyclin D1 promoter. Binding of ER α to mutant forms of promoter DNA suggest that ER α might bind to a nonconventional ERE or nonspecific promoter DNA sequence with lower affinity compared to ERE (41,42). The nonspecific binding may facilitate the binding of ER α to other transcription factors such as JUN/ATF-2 or p50/p65. Interestingly, the addition of E_2 did not increase the binding of ER α to the promoter DNA or DNA–protein complex. It appears that maximal binding of ER α has already occurred in the absence of the ligand. Presence of E_2 may cause the recruitment of coactivators such as SRC-1, leading to E_2 -induced increase in transcription (4).

It is important to note that the concentrations of spermine used in our experiments are in the physiological range. Spermidine and spermine levels were determined to be 1.15 and 0.88 mM concentration, respectively, in rat liver (59). Pera et al. (60) found putrescine, spermidine, and spermine concentrations in leukemic cells to be 0.47, 5.92, and 2.23 mM, respectively. Polyamine levels fluctuate during the cell cycle and in response to growth regulatory signals (21,22). Polyamines might also be released during histone acetylation and the unraveling of chromatin. Changes in the levels of enzymatic activities may increase the local concentration of free polyamines available for a specific function, such as activation of p38 MAP kinase and ATF-2 phosphorylation. Analysis of intracellular polyamines in our experiments demonstrated that exogenous addition of 0.1–1 mM spermine resulted in a three- to sixfold increase in spermine levels. Addition of spermine also decreased putrescine levels, probably due to feedback inhibition pathways, involving suppression of ornithine decarboxylase (ODC) activity, which is rate limiting in putrescine biosynthesis (21). Because breast tumor tissues are reported to have threefold higher levels of spermine compared to adjacent normal tissues (23), it is important to know what effect this will have on estrogenic signaling.

Our results on the effects of E_2 and spermine on endogenous cyclin D1 levels, DNA synthesis, and cell proliferation support the multifaceted role of spermine in altering cell signaling and cell growth. Effects of E_2 /spermine combinations on cyclin D1 protein levels are

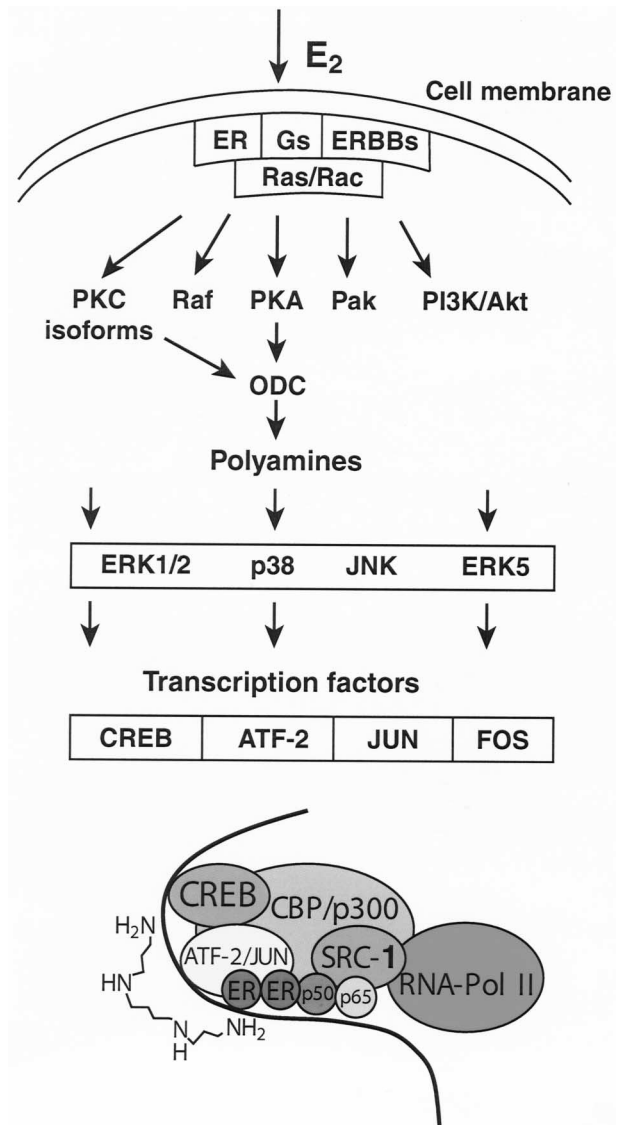


Figure 9. A model for the amplification of estrogenic signal by polyamines. Multiple kinase cascades are activated by E_2 -bound estrogen receptor (ER) in collaboration with G proteins and/or ERB (HER/neu) family of receptors. Activation of protein kinase A (PKA), protein kinase C (PKC), Raf kinases, p21 activated kinase (Pak), phosphatidylinositol 3-kinase (PI3K), and Akt have been implicated in the action of E_2 (3,51–54). Some of these kinase pathways are known to lead to the stimulation of ODC (22,64). Spermine and E_2 enhance the phosphorylation of p38 MAP kinase. Furthermore, polyamines facilitate transcriptional activation by enhancing DNA bending and protein–protein interactions (22,36). Overall, these reactions enhance cell cycle progression and cell proliferation.

moderate compared to results on the transactivation of cyclin D1 reporter gene. This is not surprising because cyclin D1 protein is under the influence of multiple post-transcriptional regulatory pathways, particularly protein degradation (61). In addition, full estrogenic response on cyclin D1 as well as the polyamine biosynthetic pathway is better observed using G1 synchronized cells (62,63). However, G1 synchronization will involve additional chemical manipulation of MCF-7 cells and hence it was avoided in this study. Thus, cyclin D1 reporter gene expression may better capture the effect of E₂ and spermine than steady-state level of cyclin D1 protein.

Figure 9 shows a model for the involvement of polyamines in the stimulation of cell signaling and cell proliferation in the presence of E₂. The amplification of cell signaling and cell growth by polyamines may occur during the activation of p38 and at the level of transcription factor binding to DNA. Furthermore, the polyamine biosynthetic enzyme, ODC, is induced by E₂ (63,64), so that endogenous polyamines are involved in these processes even in the absence of exogenous polyamines. Addition of exogenous polyamines mimics a cellular microenvironment where E₂ acts on cells in the presence of elevated polyamine levels.

In conclusion, our study demonstrates that a combination of E₂ and spermine can synergistically activate cyclin D1 at the transcriptional level. This activation is mediated primarily through the CRE/ATF motif and involves enhanced binding of ATF-2 to the ATF/CRE site. The mechanism by which E₂ and spermine facilitates DNA binding to ATF-2 involves its phosphorylation, catalyzed by the p38 MAP kinase pathway. These results indicate that ATF-2 is a critical target for polyamines in ER-mediated activation of cyclin D1. Our results provide a novel mechanism for the action of spermine in breast cancer cells and insights into the link between estrogenic and polyamine pathways in cell growth regulation.

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